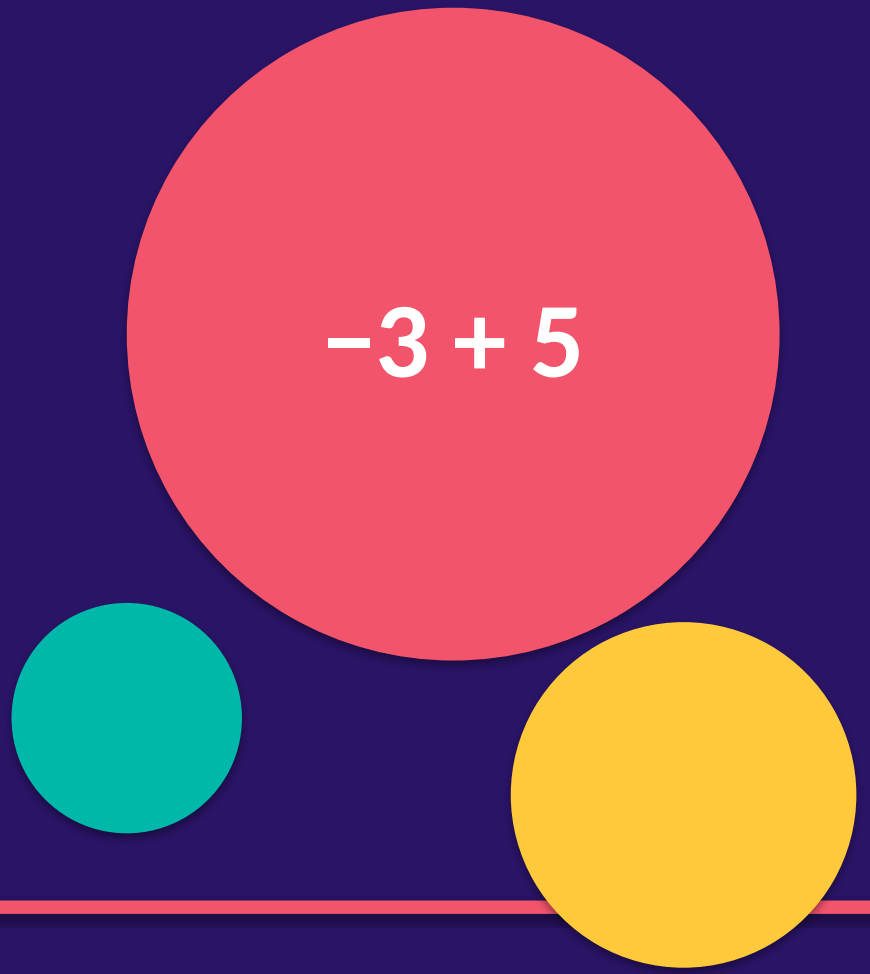
Negative numbers

Year 10 · Edexcel GCSE Foundation · 60 minutes · Mastery lesson



LO1: Order and compare negative numbers
LO2: Add and subtract with negative numbers
LO3: Multiply and divide with negative numbers

Lesson for Sienna · 60 minutes · OnMaths · Mr Ali

Have ready: mini whiteboard, pen, calculator OFF

Do Now

8 minutes

On your own. Show your working. Then we check together.

Last lesson: 23×14

Last unit: $96 \div 4$

Number skill: $5 + 3 \times 4$

Talk Task

5 minutes

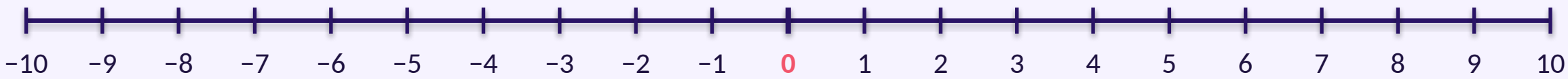
Always, Sometimes or Never?
“Subtracting always makes a number smaller.”

Say: I think it is _____ because _____ . For example, _____ .

Key words and the number line

Key facts

◀️ smaller bigger ▶️

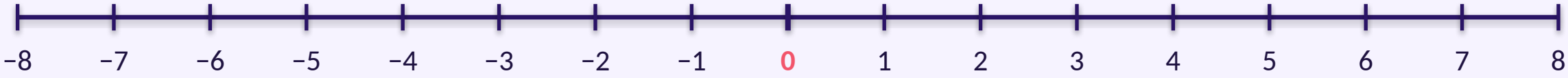


Negative number	less than zero, e.g. −4
Positive number	more than zero, e.g. 4
Integer	a whole number: … −2, −1, 0, 1, 2 …
Number line	numbers get BIGGER as you move RIGHT

LO1 I do: order negative numbers

I do

Put in order, smallest first: 3, −2, 5, −7, 0



LO1 We do

We do

You say the answer first, then click to check.

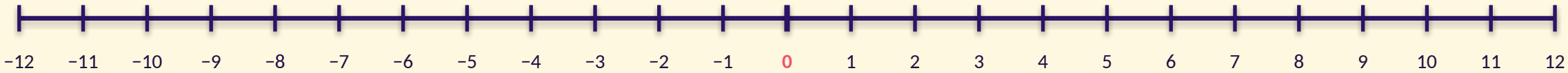
−2 ☐ 3 (< or >)

−6 ☐ −1 (< or >)

0 ☐ −5 (< or >)

Coldest: −4°C, 2°C, −8°C

Use the number line:



LO1 Mini whiteboards (1 of 2)

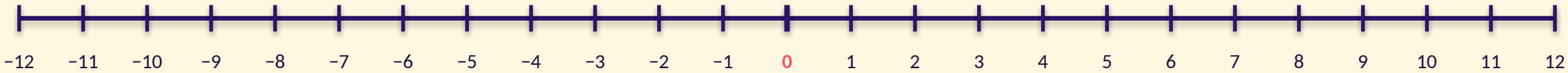
MWB

Write your answer, show me, then click.

Order, smallest first: −1, −6, 4, −3

Which is bigger: −5 or −2?

Use the number line:



LO1 Mini whiteboards (2 of 2)

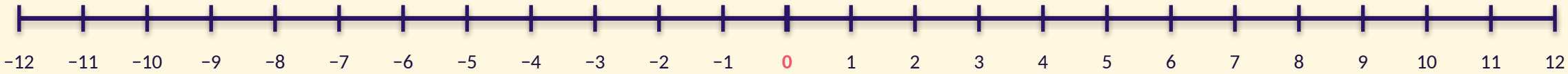
MWB

Write your answer, show me, then click.

−10 ☐ −12 (< or >)

Which is warmer: −3°C or −7°C?

Use the number line:



LO1 You do

B · S · G · P

BRONZE

Order, smallest first:
3, −2, 0, −5

SILVER

Write < or >:
−8 ☐ −3 −1 ☐ −11

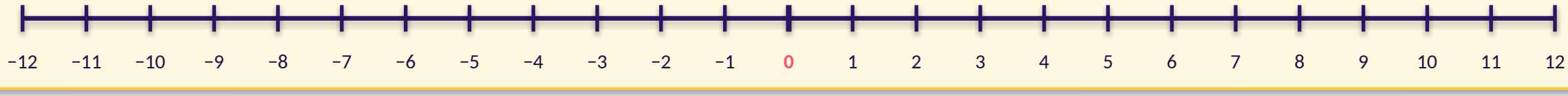
GOLD

6am temperatures: Leeds −2°C, Oslo −9°C,
Rome 4°C, Moscow −12°C.
Which city is coldest? Which is warmest?

PLATINUM

Write down a number between −4 and −3.5.
Explain how you know.

Use the number line:



LO2 I do: add and subtract on a number line

I do

+ move RIGHT **− move LEFT**





LO2 I do: two signs together

I do

Two signs next to each other:
SAME signs → + DIFFERENT signs → −

$-2 - (-6)$

$3 + (-5)$

$-1 - (+4)$

LO2 More examples (optional)

I do ×3 · optional

$-6 + 4$

$-1 - 5$

$7 - (-3)$

LO2 We do

We do

Use the number line in your head (or draw one).

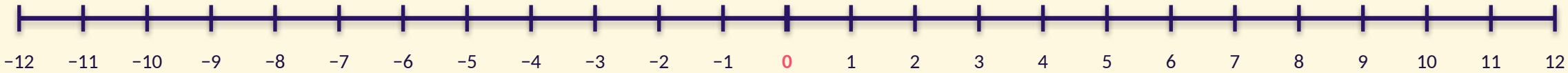
It is −4°C. The temperature rises 9 degrees.

It is 3°C. The temperature falls 8 degrees.

−2 − 4

5 + (−8)

Use the number line:



LO2 Mini whiteboards (1 of 2)

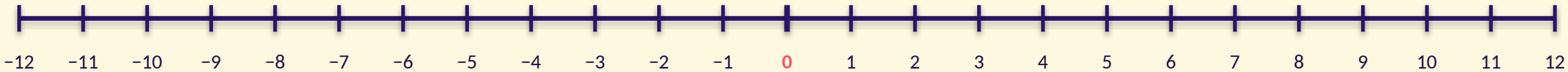
MWB

Write your answer, show me, then click.

$-5 + 3$

$2 - 9$

Use the number line:



LO2 Mini whiteboards (2 of 2)

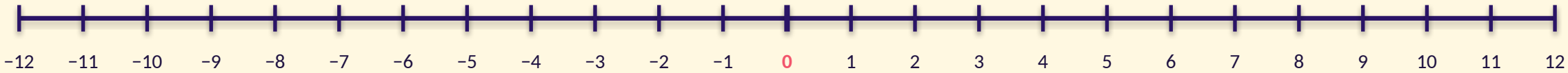
MWB

Write your answer, show me, then click.

$-4 - (-7)$

$-6 + (-2)$

Use the number line:



LO2 You do

B · S · G · P

BRONZE

a) −7 + 4
b) 6 − 9

SILVER

a) −2 − 5
b) −8 + 11

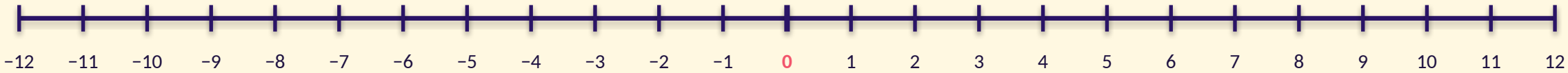
GOLD

a) 4 − (−6)
b) −5 + (−3)

PLATINUM

Oslo is −6°C. London is 9 degrees warmer.
Rome is 5 degrees warmer than London.
What is the temperature in Rome?

Use the number line:



LO3 I do: multiply and divide

I do

SAME signs → POSITIVE

+ × + = + **− × − = +**

DIFFERENT signs → NEGATIVE

+ × − = − **− × + = −**

Step 1: ignore the signs and work out the number.
Step 2: same signs → +, different signs → −.
The same rules work for ÷.

−3 × 4

−5 × −2

−20 ÷ 4

−18 ÷ −3

LO3 We do

We do

Step 1: ignore the signs. Step 2: same signs +, different signs −.

6×-3

-7×-2

$24 \div -6$

$-30 \div -5$

Help: SAME signs → POSITIVE (− × − = +) DIFFERENT signs → NEGATIVE (− × + = −)

LO3 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

-4×5

-9×-3

Help: SAME signs → POSITIVE (− × − = +) DIFFERENT signs → NEGATIVE (− × + = −)

LO3 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

$-36 \div 9$

$(-2)^2$

Help: SAME signs → POSITIVE (− × − = +) DIFFERENT signs → NEGATIVE (− × + = −)

LO3 You do

B · S · G · P

BRONZE

a) -2×6
b) 3×-5

SILVER

a) -8×-4
b) $-40 \div 8$

GOLD

a) $-48 \div -6$
b) $(-3)^2$

PLATINUM

Find two integers that multiply to −12 and add to 1.

Help: SAME signs → POSITIVE (− × − = +) DIFFERENT signs → NEGATIVE (− × + = −)

Exam questions

Edexcel style

1. At midnight it was −3°C. At midday it was 7°C.
   By how many degrees did the temperature rise?

(1 mark)

2. Work out $-4 \times -5 + 3$

(2 marks)

3. Ali's bank balance is −£25. He pays in £60.
   What is his new balance?

(1 mark)

Exit ticket

5 minutes

On your own. Also on your OnMaths page, where it is marked for you.

1. It is −2°C. The temperature falls by 5 degrees.
   What is the new temperature?

2. Work out $4 - (-6)$

3. Work out -6×-4

Finished early? Extension

Extension

A diver is at −12 m. She rises 5 m, then dives down 8 m.
Where is she now?

Fill the gap: $-4 \times \square = 28$

Which is greater: $(-5)^2$ or -5^2? Explain.

Homework

Due next Friday

5 worded questions on negative numbers (9 marks).

Do it online on your OnMaths page – it marks itself.
Show your working on paper.

Stuck? Draw a number line.

Today I learned: order negative numbers ✓ add and subtract ✓ multiply and divide ✓